Reinforcement Learning

from one delivery van to the Bellman equation

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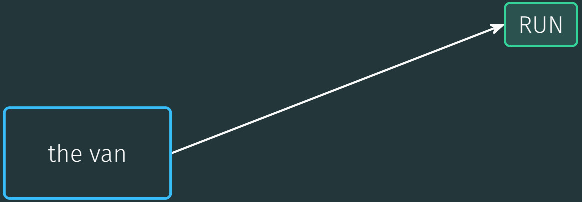
Motivation

One van, one call a week

every week: exactly one call

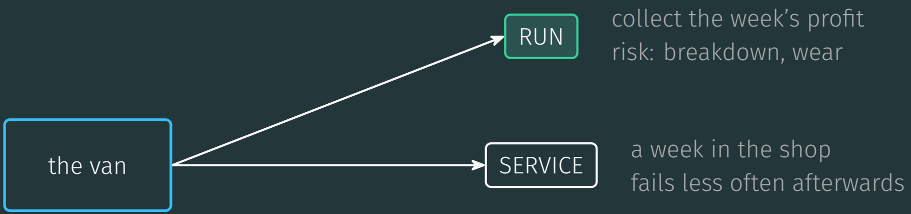
One van, one call a week



collect the week’s profit
risk: breakdown, wear

every week: exactly one call

One van, one call a week



every week: exactly one call

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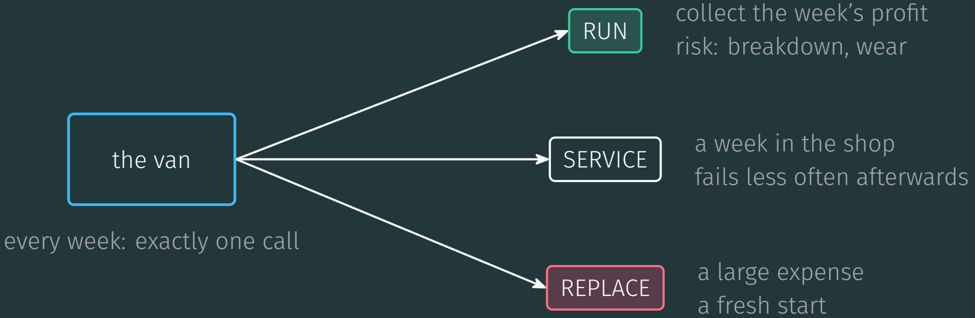
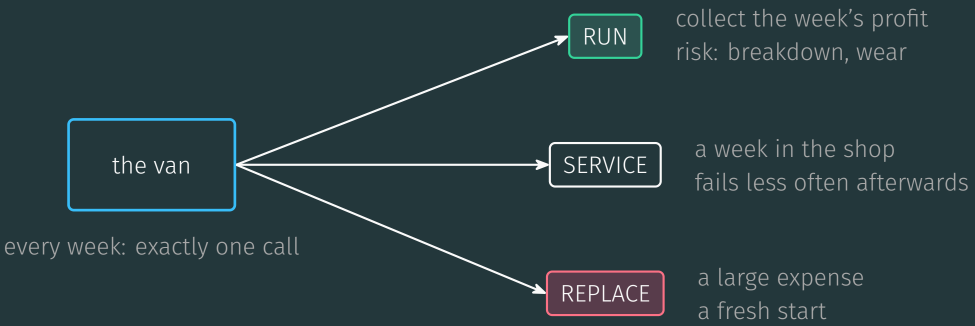


One van, one call a week



short-term cost against long-term payoff, chance at every step

Markov decision processes

The model: four ingredients

S: states

four wear levels

The model: four ingredients

S: states

four wear levels

A: actions

three calls

The model: four ingredients

S: states

four wear levels

A: actions

three calls

T: dynamics

how the world reacts

The model: four ingredients

S: states	A: actions	T: dynamics	$\gamma = 0.9$
four wear levels	three calls	how the world reacts	how much the future counts

The model: four ingredients

S: states	A: actions	T: dynamics	$\gamma = 0.9$
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$$(s', r) = T(s, a)$$

The model: four ingredients

S: states	A: actions	T: dynamics	$\gamma = 0.9$
four wear levels	three calls	how the world reacts	how much the future counts

$$(\underbrace{s'}_{\text{the state next week}}, \underbrace{r}_{\text{the reward}}) = T(s, a)$$

The fleet sheet

HEALTHY

RUN profit	+95
breakdown	2%

The fleet sheet

HEALTHY $\xrightarrow{\text{wear slips}}$ WORN

	HEALTHY	WORN
RUN profit	+95	+72
breakdown	2%	8%

The fleet sheet

HEALTHY $\xrightarrow{\text{wear slips}}$ WORN $\rightarrow$ SHAKY

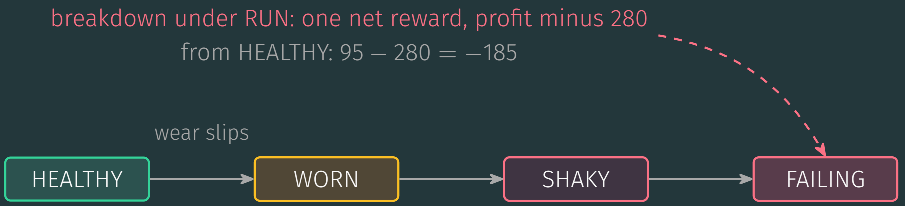
	HEALTHY	WORN	SHAKY
RUN profit	+95	+72	+40
breakdown	2%	8%	28%

The fleet sheet

wear slips

	HEALTHY →	WORN →	SHAKY →	FAILING
RUN profit	+95	+72	+40	+16
breakdown	2%	8%	28%	55%

The fleet sheet



	HEALTHY	WORN	SHAKY	FAILING
RUN profit	+95	+72	+40	+16
breakdown	2%	8%	28%	55%

The fleet sheet

breakdown under RUN: one net reward, profit minus 280

from HEALTHY: $95 - 280 = -185$

wear slips

HEALTHY → WORN → SHAKY → FAILING

	HEALTHY	WORN	SHAKY	FAILING
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SERVICE: −50, a week in the shop (WORN back to HEALTHY 95% of the time)

The fleet sheet

breakdown under RUN: one net reward, profit minus 280

from HEALTHY: $95 - 280 = -185$

wear slips

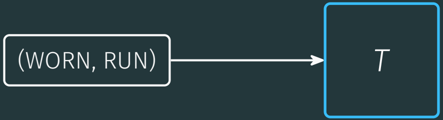
HEALTHY → WORN → SHAKY → FAILING

	HEALTHY	WORN	SHAKY	FAILING
RUN profit	$+95$	$+72$	$+40$	$+16$
breakdown	2%	8%	28%	55%

SERVICE: −50, a week in the shop (WORN back to HEALTHY 95% of the time)

REPLACE: −130, a brand-new HEALTHY van next week

Same pair in, different week out



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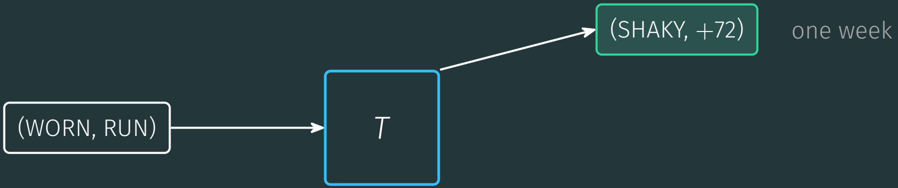

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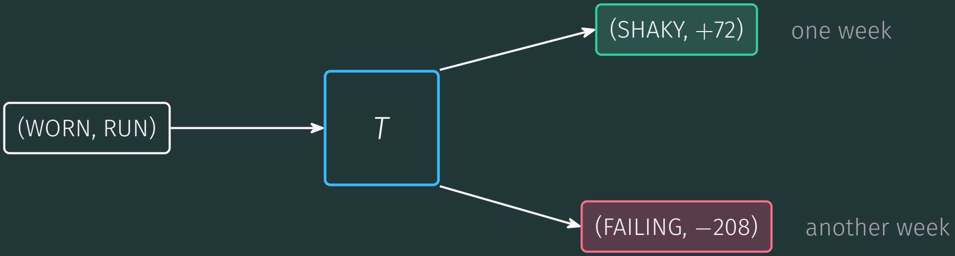

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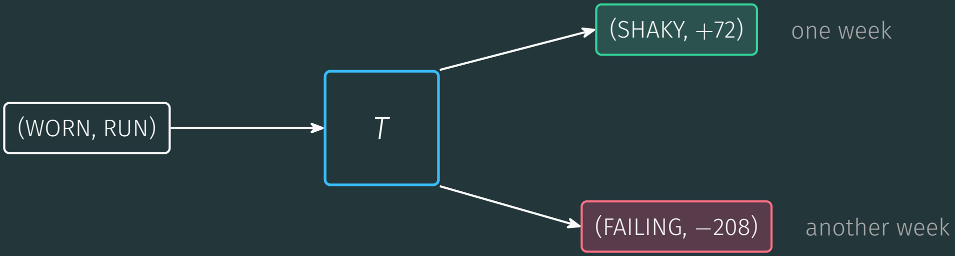

same input, different output: the dynamics T is random

Policies and trajectories

Playbook in, tape out

a policy (the playbook): $\pi : S \to A$, one call per state

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a policy (the playbook): $\pi : S \to A$, one call per state

HEALTHY $\to$ RUN | WORN $\to$ SERVICE | SHAKY $\to$ REPLACE | FAILING $\to$ REPLACE

each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$

Playbook in, tape out

a policy (the playbook): $\pi : S \to A$, one call per state

HEALTHY $\to$ RUN	WORN $\to$ SERVICE	SHAKY $\to$ REPLACE	FAILING $\to$ REPLACE

each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$

wk 1
HEALTHY, RUN
$+95$

Playbook in, tape out

a policy (the playbook): $\pi : S \to A$, one call per state

HEALTHY $\to$ RUN	WORN $\to$ SERVICE	SHAKY $\to$ REPLACE	FAILING $\to$ REPLACE

each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$

$$\boxed{\begin{array}{c}\text{wk 1}\\ \text{HEALTHY, RUN}\\ +95\end{array}} \longrightarrow \boxed{\begin{array}{c}\text{wk 2}\\ \text{WORN, SERVICE}\\ -50\end{array}}$$

Playbook in, tape out

a policy (the playbook): $\pi : S \to A$, one call per state

HEALTHY → RUN | WORN → SERVICE | SHAKY → REPLACE | FAILING → REPLACE

each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$

$$\boxed{\begin{array}{c}\text{wk 1}\\ \text{HEALTHY, RUN}\\ +95\end{array}} \longrightarrow \boxed{\begin{array}{c}\text{wk 2}\\ \text{WORN, SERVICE}\\ -50\end{array}} \longrightarrow \boxed{\begin{array}{c}\text{wk 3}\\ \text{HEALTHY, RUN}\\ +95\end{array}}$$

Playbook in, tape out

a policy (the playbook): $\pi : S \to A$, one call per state

HEALTHY $\to$ RUN | WORN $\to$ SERVICE | SHAKY $\to$ REPLACE | FAILING $\to$ REPLACE

each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$

wk 1		wk 2		wk 3		wk 4
HEALTHY, RUN	$\to$	WORN, SERVICE	$\to$	HEALTHY, RUN	$\to$	HEALTHY, RUN
$+95$		-50		$+95$		$+95$

Playbook in, tape out

a policy (the playbook): $\pi : S \to A$, one call per state

HEALTHY $\to$ RUN

WORN $\to$ SERVICE

SHAKY $\to$ REPLACE

FAILING $\to$ REPLACE

each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$

wk 1
HEALTHY, RUN
$+95$ → wk 2
WORN, SERVICE
-50 → wk 3
HEALTHY, RUN
$+95$ → wk 4
HEALTHY, RUN
$+95$ → wk 5
WORN

the trajectory: $s_1, a_1, r_1,\ s_2, a_2, r_2,\ \ldots$ one week per step, forever

Playbook in, tape out

a policy (the playbook): $\pi : S \to A$, one call per state

HEALTHY $\to$ RUN | WORN $\to$ SERVICE | SHAKY $\to$ REPLACE | FAILING $\to$ REPLACE

each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$

wk 1 HEALTHY, RUN $+95$ $\to$ wk 2 WORN, SERVICE -50 $\to$ wk 3 HEALTHY, RUN $+95$ $\to$ wk 4 HEALTHY, RUN $+95$ $\to$ wk 5 WORN

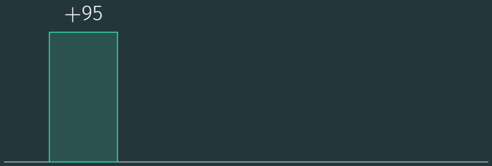
the trajectory: $s_1, a_1, r_1,\; s_2, a_2, r_2,\; \ldots$ one week per step, forever

the tape is random; another run, same start: (HEALTHY, RUN, -185) $\to$ FAILING

The goal: accumulating reward

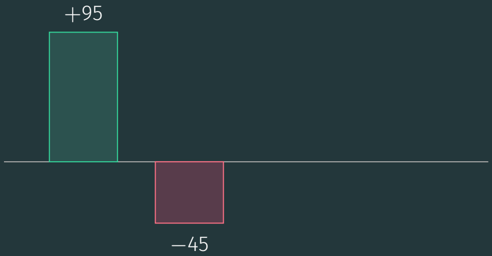
The return: one number per tape

$$G_1 = r_1$$



The return: one number per tape

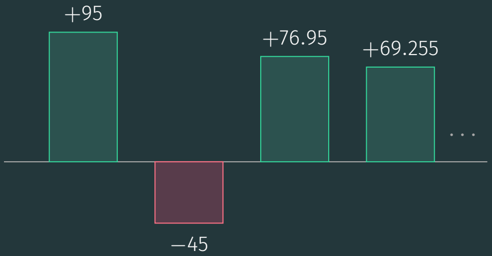
$$G_1 = \underbrace{r_1}_{\text{in full}} + \underbrace{\gamma\, r_2}_{\text{at 90\%}}$$



each week shrunk
by $\gamma = 0.9$

The return: one number per tape

$$G_1 = \underbrace{r_1}_{\text{in full}} + \underbrace{\gamma\, r_2}_{\text{at 90\%}} + \underbrace{\gamma^2 r_3}_{\text{at 81\%}} + \cdots$$



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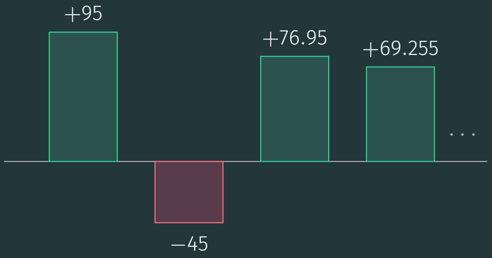
$$G_1 = \underbrace{r_1}_{\text{in full}} + \underbrace{\gamma\, r_2}_{\text{at 90\%}} + \underbrace{\gamma^2 r_3}_{\text{at 81\%}} + \cdots$$

$$G_1 = 95 + 0.9\,(-50) + 0.81 \cdot 95 + 0.729 \cdot 95 + \cdots = 95 - 45 + 76.95 + 69.255 + \cdots$$



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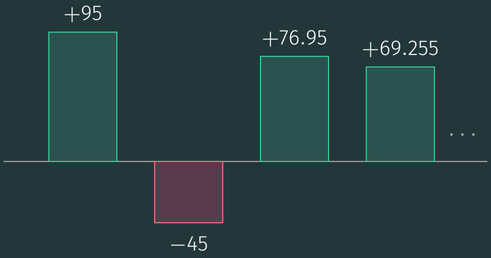


each week shrunk
by $\gamma = 0.9$

the shrinking factors keep
the never-ending sum finite

The recurrence

$$G_i = r_i + \gamma\, r_{i+1} + \gamma^2 r_{i+2} + \cdots$$

r_i	r_{i+1}	r_{i+2}	$\cdots$

The recurrence

$$G_i = r_i + \gamma \left(r_{i+1} + \gamma\, r_{i+2} + \cdots \right)$$

$$\boxed{r_i} \qquad \boxed{\boxed{r_{i+1}} \quad \boxed{r_{i+2}} \quad \boxed{\cdots}}$$

G_{i+1}, discounted once by γ

The recurrence

$$G_i = \underbrace{r_i}_{\text{this week}} + \gamma \underbrace{G_{i+1}}_{\text{everything after}}$$

r_i | r_{i+1} | r_{i+2} | $\cdots$

G_{i+1}, discounted once by γ

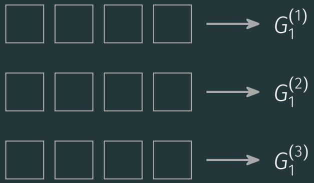
The recurrence

$$G_i = \underbrace{r_i}_{\text{this week}} + \gamma \underbrace{G_{i+1}}_{\text{everything after}}$$

G_{i+1}, discounted once by γ

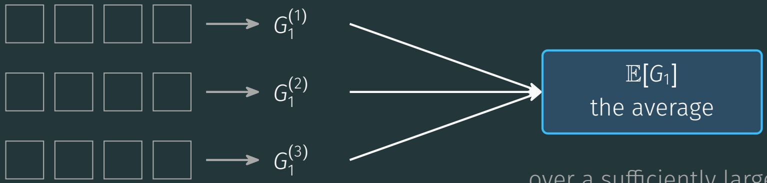
this week's reward $+ \gamma \times$ all the rest

Maximize the average



same playbook, same start: a different G_1 every run

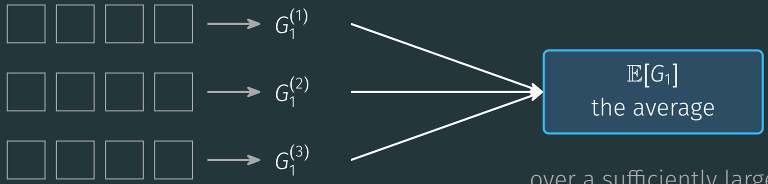
Maximize the average



over a sufficiently large
set of trajectories

same playbook, same start: a different G_1 every run

Maximize the average



one plain number

over a sufficiently large set of trajectories

same playbook, same start: a different G_1 every run

Maximize the average

one plain number

over a sufficiently large set of trajectories

same playbook, same start: a different G_1 every run

the goal: find the policy π that maximizes $\mathbb{E}[G_1]$

The Q-function and the Bellman equation

The scorecard $Q^\star$

$$Q^\star(s, a) = \mathbb{E}\left[\, G_i \;\middle|\; s_i = s,\ a_i = a \,\right]$$

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	RUN	SERVICE	REPLACE
HEALTHY	311.0	229.9	149.9

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	RUN	SERVICE	REPLACE
HEALTHY	311.0	229.9	149.9
WORN	203.4	226.1	149.9

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	RUN	SERVICE	REPLACE
HEALTHY	311.0	229.9	149.9
WORN	203.4	226.1	149.9
SHAKY	96.5	126.4	149.9

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	RUN	SERVICE	REPLACE
HEALTHY	311.0	229.9	149.9
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SHAKY	96.5	126.4	149.9
FAILING	−3.1	86.9	149.9

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$\pi^\star$: read each row, keep the call with the largest value

	RUN	SERVICE	REPLACE
HEALTHY	$\boxed{311.0}$	229.9	149.9
WORN	203.4	$\boxed{226.1}$	149.9
SHAKY	96.5	126.4	$\boxed{149.9}$
FAILING	−3.1	86.9	$\boxed{149.9}$

row by row: the kept call

The scorecard $Q^\star$

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SHAKY	96.5	126.4	149.9
FAILING	−3.1	86.9	149.9

row by row: the kept call

149.9 beats 126.4 by 23.5

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SHAKY	96.5	126.4	$\boxed{149.9}$
FAILING	−3.1	86.9	$\boxed{149.9}$

row by row: the kept call

149.9 beats 126.4 by 23.5

the optimal playbook replaces a van that still starts every morning

The Bellman equation

$$G_i = r_i + \gamma\, G_{i+1} \qquad \text{(every single week)}$$

$$\boxed{(s_i,\ a_i)} \xrightarrow{\text{pays } r_i} \boxed{(s_{i+1},\ a_{i+1})} \qquad \text{counted at } \gamma$$

The Bellman equation

$$Q^\star(s_i, a_i) = \mathbb{E}\left[\, r_i + \gamma\, G_{i+1} \,\right]$$ (average both sides)

$$(s_i,\ a_i) \xrightarrow{\text{pays } r_i} (s_{i+1},\ a_{i+1})$$ counted at γ

The Bellman equation

$$\boxed{Q^\star(s_i, a_i) = \mathbb{E}\Big[\underbrace{r_i}_{\text{this week's cash}} + \gamma \underbrace{Q^\star(s_{i+1}, a_{i+1})}_{\text{next week's value}} \Big]}$$

replacing G_{i+1} by its average $Q^\star(s_{i+1}, a_{i+1})$: averaging in two rounds

$$\boxed{(s_i,\ a_i)} \xrightarrow{\text{pays } r_i} \boxed{(s_{i+1},\ a_{i+1})} \qquad \text{counted at } \gamma$$

The Bellman equation

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replacing G_{i+1} by its average $Q^\star(s_{i+1}, a_{i+1})$: averaging in two rounds

$$(s_i,\ a_i) \xrightarrow{\text{pays } r_i} (s_{i+1},\ a_{i+1}) \qquad \text{counted at } \gamma$$

one equation per cell: twelve equations pin down the table

SARSA: learning from the logbook

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72
Overwriter card = last week	72
Filing cabinet true average of every week	72
Nudger move 15% toward the week	72

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72
Overwriter card = last week	72	72
Filing cabinet true average of every week	72	72
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Overwriter card = last week	72	72	72
Filing cabinet true average of every week	72	72	72
Nudger move 15% toward the week	72	72	72

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72	−208
Overwriter card = last week	72	72	72	−208
Filing cabinet true average of every week	72	72	72	2
Nudger move 15% toward the week	72	72	72	30

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72	−208	+72
Overwriter card = last week	72	72	72	−208	72
Filing cabinet true average of every week	72	72	72	2	16
Nudger move 15% toward the week	72	72	72	30	36.3

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72	−208	+72	
Overwriter card = last week	72	72	72	−208	72	hysterical
Filing cabinet true average of every week	72	72	72	2	16	right, but stores every card
Nudger move 15% toward the week	72	72	72	30	36.3	calm, and no cabinet

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72	−208	+72	
Overwriter card = last week	72	72	72	−208	72	hysterical
Filing cabinet true average of every week	72	72	72	2	16	right, but stores every card
Nudger move 15% toward the week	72	72	72	30	36.3	calm, and no cabinet

$$q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha \left[r_i - q(s_i, a_i) \right]$$

(the Nudger, written down)

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72	−208	+72	
Overwriter card = last week	72	72	72	−208	72	hysterical
Filing cabinet true average of every week	72	72	72	2	16	right, but stores every card
Nudger move 15% toward the week	72	72	72	30	36.3	calm, and no cabinet

$$q(s_i, a_i) \leftarrow \underbrace{q(s_i, a_i)}_{\text{the card}} + \underbrace{\alpha}_{0.15} \underbrace{\left[r_i - q(s_i, a_i) \right]}_{\text{the surprise}}$$

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72	−208	+72	
Overwriter card = last week	72	72	72	−208	72	hysterical
Filing cabinet true average of every week	72	72	72	2	16	right, but stores every card
Nudger move 15% toward the week	72	72	72	30	36.3	calm, and no cabinet

$$q(s_i, a_i) \leftarrow \underbrace{q(s_i, a_i)}_{\text{the card}} + \underbrace{\alpha}_{0.15} \underbrace{\left[r_i - q(s_i, a_i) \right]}_{\text{the surprise}}$$

nudge, never overwrite: one bad week must not erase a year

The alpha knob

same five weeks, three step sizes

The alpha knob

same five weeks, three step sizes

$\boxed{\alpha = 0.9}$

$$\boxed{72} \xrightarrow{\text{the } -208 \text{ week lands}} \boxed{-180} \xrightarrow{\text{then } +72 \text{ lands}} \boxed{46.8}$$

whiplash

The alpha knob

same five weeks, three step sizes

$\alpha = 0.9$

72

the −208 week lands

−180

then +72 lands

46.8

whiplash

$\alpha = 0.05$

72

58

58.7

barely notices

The alpha knob

same five weeks, three step sizes

	$\alpha = 0.9$	$\alpha = 0.15$	$\alpha = 0.05$
	72	72	72
the -208 week lands	-180	30	58
then $+72$ lands	46.8	36.3	58.7
	whiplash	the calm middle	barely notices

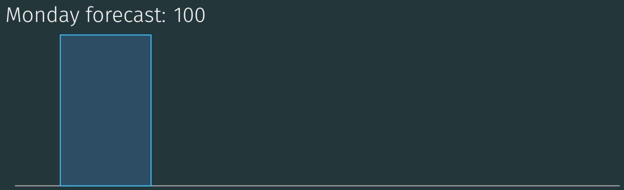
The alpha knob

same five weeks, three step sizes

	$\alpha = 0.9$	$\alpha = 0.15$	$\alpha = 0.05$
	72	72	72
the −208 week lands	−180	30	58
then +72 lands	46.8	36.3	58.7
	whiplash	the calm middle	barely notices

small steady steps, never overwrite

The planner’s Monday



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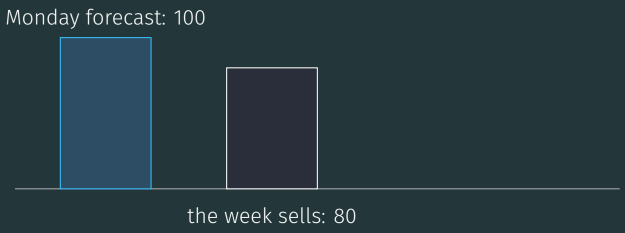

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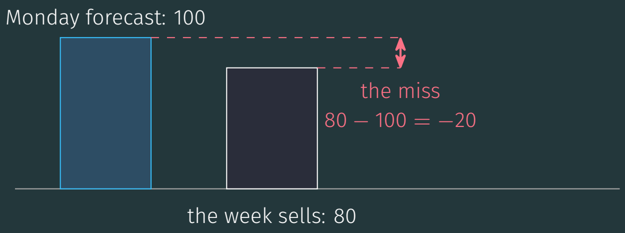

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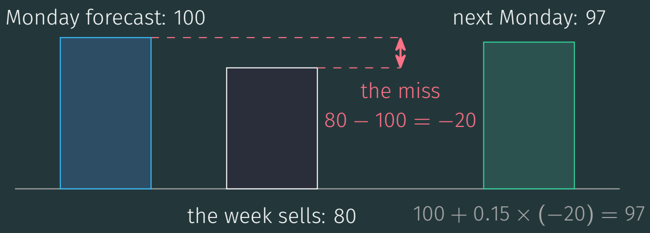

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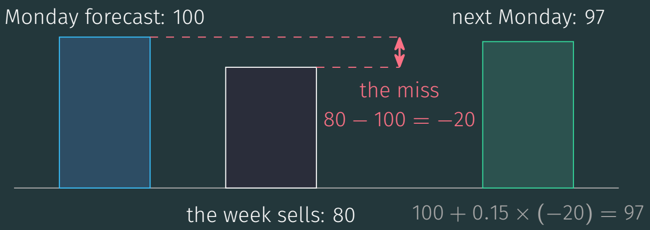
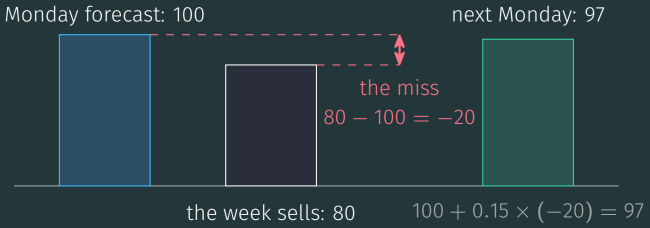


$$\text{new forecast} = \text{old forecast} + 0.15 \times \text{miss}$$

The planner’s Monday



$$\text{new forecast} = \underbrace{\text{old forecast}}_{q(s_i,\, a_i)} + \underbrace{0.15}_{\alpha} \times \underbrace{\text{miss}}_{r_i - q(s_i, a_i)}$$

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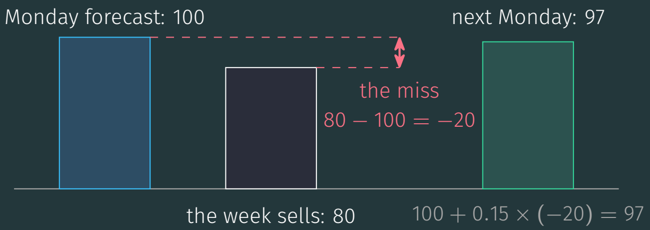

the planner’s name:

exponential smoothing

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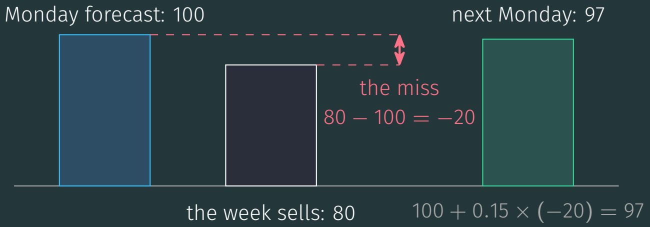

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SARSA forecasts the value of a call the way a planner forecasts sales

The cash-only ledger lies

the cash-only ledger

REPLACE: —130, every single week

worst call on the sheet

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SERVICE: —50

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the scorecard $Q^\star$ you saw

REPLACE crowned at SHAKY and FAILING: 149.9

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what does the cash-only ledger miss?

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one week → the same cash → a HEALTHY van left in the lot

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what does the cash-only ledger miss?

one week → the same cash → a HEALTHY van left in the lot

one week → the same cash → a FAILING van left in the lot

the week leaves you a van: same cash, different futures

Grade the week like a trade-in

S	A	R	S	A	
s_i	a_i	r_i	s_{i+1}	a_{i+1}	one logbook line

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		cash	the pair you land on		

trade-in price: $\gamma \times q(s_{i+1}, a_{i+1})$

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trade-in price: $\gamma \times q(s_{i+1}, a_{i+1})$

$$\text{the week's worth} = \underbrace{r_i}_{\text{cash}} + \gamma \underbrace{q(s_{i+1}, a_{i+1})}_{\text{trade-in, at your own price}}$$

priced from your own card wall: $q(s_{i+1}, a_{i+1})$ stands in for the unknown $Q^\star$ (the NPV move)

Grade the week like a trade-in

S A R S A

s_i a_i r_i s_{i+1} a_{i+1} one logbook line

cash

the pair you land on

trade-in price: $\gamma \times q(s_{i+1}, a_{i+1})$

$$q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha \Big[r_i - q(s_i, a_i) \Big] \qquad \text{(cash only)}$$

priced from your own card wall: $q(s_{i+1}, a_{i+1})$ stands in for the unknown $Q^\star$ (the NPV move)

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$$q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha \left[\underbrace{r_i + \gamma\, q(s_{i+1}, a_{i+1})}_{\text{the actual, grown up}} - q(s_i, a_i) \right]$$

priced from your own card wall: $q(s_{i+1}, a_{i+1})$ stands in for the unknown $Q^\star$ (the NPV move)

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$\alpha = 0.15$, constant · $\varepsilon = 0.15$: a random call in 15% of weeks · a_{i+1}: actually executed, never a max

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Bellman, with the expectation replaced by an arriving average

One week, one nudge

twelve guesses, all start at 0

	RUN	SERVICE	REPLACE
HEALTHY	0	0	0
WORN	0	0	0
SHAKY	0	0	0
FAILING	0	0	0

One week, one nudge

week one in the logbook: (HEALTHY, RUN, +95, WORN, SERVICE)

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week one in the logbook: (HEALTHY, RUN, +95, WORN, SERVICE)

$$\text{evidence} = r_i + \gamma\, q(\text{WORN}, \text{SERVICE}) = 95 + 0.9 \times 0 = 95$$

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$$\text{surprise} = 95 - 0 = 95$$

nudge: $+\,0.15 \times 95$

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nudge: $+\,0.15 \times 95$

twelve guesses, all start at 0

one cell moves, eleven stay put

	RUN	SERVICE	REPLACE
HEALTHY	14.25	0	0
WORN	0	0	0
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FAILING	0	0	0

One week, one nudge

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a few hundred weeks later:
the three bands emerge

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the cells settle below $Q^\star$ and keep jittering, but each row keeps the right call

The whole story

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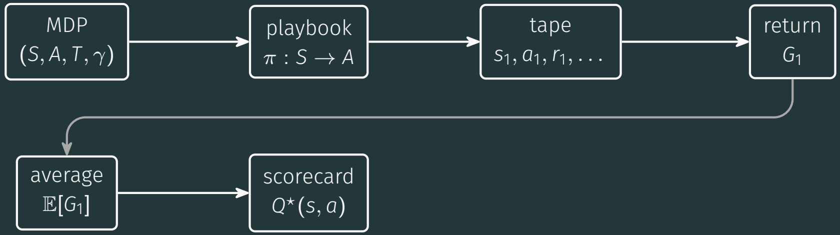
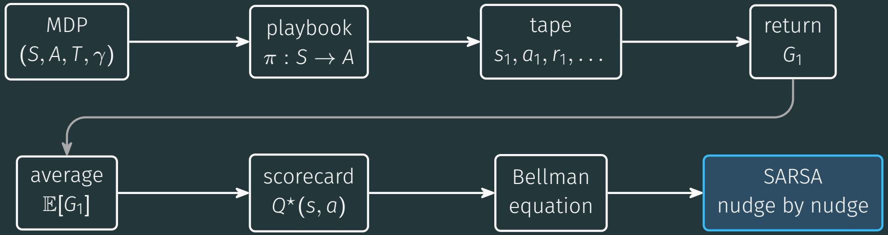


The whole story



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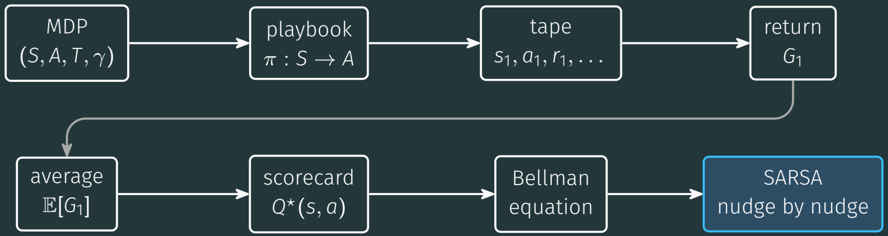
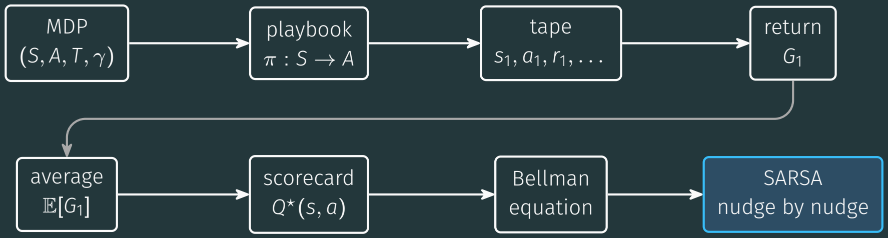


SARSA solves the Bellman equation by averaging evidence instead of computing expectations

The whole story



SARSA solves the Bellman equation by averaging evidence instead of computing expectations

play it: `https://sml-fs26.github.io/classic_rl/repair-or-replace/`

Appendix: why the nudge works

The Robbins-Monro iteration

find the number x^* that equals the average of noisy samples, one at a time

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$$x \leftarrow x + \alpha_n \,(\text{sample} - x)$$

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SARSA is Robbins-Monro on the Bellman equation, cell by cell

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A_n: the average of the first n observations $x_1, \ldots, x_n$

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$$A_n = \underbrace{A_{n-1}}_{\text{average so far}} + \underbrace{\frac{1}{n}\left(x_n - A_{n-1}\right)}_{\text{one share of the surprise}}$$

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SARSA is a running average of the weekly evidence